



Examiners' Report Lead Examiner Feedback

June 2023

Pearson BTEC Nationals
In Applied Science/Forensic and Criminal
Investigation (31617H)
Unit 1: Principles and Applications of Science I

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Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this [link](#)

Introduction

Biology

Structure and functions of cells and tissues.

Medical professionals need to understand the structure and workings of cells. They build on this knowledge to understand how the body stays healthy as well as the symptoms and causes of some diseases. This allows them to diagnose and treat illnesses. The study of bacterial prokaryotic cells gives an understanding of how some other diseases are caused and can be treated.

Chemistry

This paper is one of three externally assessed papers worth 30 marks each that assess Unit 1 – Principles and Applications of Science I. This paper covers key chemistry concepts including atomic structure and bonding and chemical and physical properties of substances to their uses.

Physics

The examination was designed to allow learners to show their knowledge and understanding of 'Waves in Communication'. The examination was comparable to those set previously as there were no changes to the specification or the composition of the paper.

The topics included transverse and longitudinal waves, stationary waves and the application of these waves. The applications included the transmission of signals by mobile phones and the passage of light waves through optical fibres in an endoscope.

Overall Performance of the Unit

Biology

Learners should show that they understand the relationship between the structure and function of cells and tissues throughout the unit. Learners still struggle to recall definitions and showed a lack of understanding of cell theory- the relationship between cells, tissues, organs and organ systems.

Learners were able to perform the given calculations using clear and logical working.

Learners had been asked previously to describe and explain the sliding muscle filament theory, this series the learners were asked to demonstrate their understanding of muscle contraction with a line drawing. Most learners indicated that the muscle filament would be shorter, but fewer learners indicated that there would therefore be more overlap between the actin and myosin.

The majority of learners were able to access the extended open response question and gain at least one mark in level 1 for their knowledge of atherosclerosis.

Chemistry

Learners that were successful this year showed a good understanding of key terms and scientific vocabulary and were able to apply this understanding to given scenarios. They were precise with their language and explained in detail where necessary. Learners that carried out their calculations in a clear and logical manner helped them to arrive at the correct answer.

Physics

Learners were generally able to analyse graphs to find amplitude and period and were able to use the period to determine the frequency of a wave.

Recalling specific information in the description of both longitudinal and transverse waves was found to be more difficult.

Describing how a stationary wave is produced also presented some difficulties although learners were able to recognise nodes, antinodes and harmonics from a diagram of a standing wave.

Many more learners have developed skills in the use of calculators and were able to use their calculators to deal with complex equations, although the use of squares and square roots still presented problems to some learners.

Although mobile phones are in everyday use, the method of transmission of signals from one phone to another was not clearly understood. Similarly,

although many learners use 'Bluetooth®' headphones their knowledge of the waves transmitted by this low power device was quite limited.

The medical application of the use of optical fibres in endoscopes required learners to have knowledge of the structure of the endoscope and to understand the method by which light travels up and down the optical fibres. Those learners that used a labelled diagram to show total internal reflection were able to describe the passage of light through the fibres more accurately. However, the majority of learners found it difficult to describe the mechanism used in the endoscope for an image to be obtained.

Individual Questions

Biology

Question 1

1(a) Most learners were able to achieve at least one mark for this question, generally for 'system'. Those that did not score a mark chose 'organ' again, or 'organelle', and there was evidence that a significant number of students did not know the difference between the two. Many correctly identified the tissue, although again 'organelle' was a common incorrect answer. Some learners named the tissue and organ system in the figure rather than giving the generic terms, but these were accepted as correctly names examples.

This response was awarded two marks. For circle the learner has given a correct response, one mark was given. For the diamond, 'system' alone was accepted and another mark given.

'Organ system' or any named organ system were also creditworthy. Learners were not being tested on if they recognised the digestive system from a diagram but on cell theory.

1 Figure 1 shows the five levels of organisation in a multicellular organism.

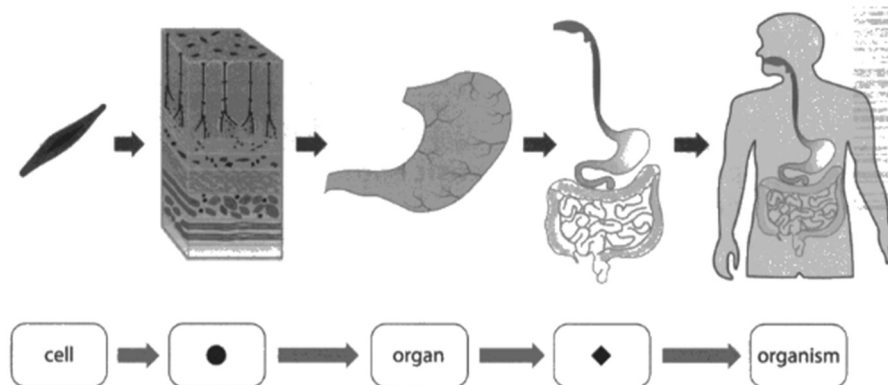


Figure 1

(a) Identify the missing words represented by the shapes ● and ◆ in Figure 1.

(2)

● Tissue
◆ ~~Tissue~~ system

The following response was awarded one mark. For the circle, the learner has given a named example of a tissue and therefore awarded one mark, in line with the additional guidance on the mark scheme.

(a) Identify the missing words represented by the shapes ● and ◆ in Figure 1.

(2)

● muscle
◆

1(b)(i) The vast majority of these answers were calculated correctly. Observed errors included inverting the calculation or multiplying the numbers together.

This response was awarded two marks. The answer did not have to be given on the answer line. The learner's final answer is 4, which was correct and so two marks were awarded.

(i) The objective lens magnification of a light microscope can be calculated by:

$$\text{objective lens magnification} = \frac{\text{magnifying power}}{\text{eye piece lens magnification}}$$

Calculate the objective lens magnification of the light microscope with a × 10 eye piece lens magnification and a × 40 magnifying power.

(2)

Show your working.

$$Olm = \frac{mp}{epm} \rightarrow Olm = \frac{40}{10} \rightarrow 4$$

objective lens magnification = ×

The following response was awarded one mark. The learner has used our values and our equation, but it is inverted.

The answer of 0.25 was a common error and was awarded one mark for ECF for a correct evaluation from an incorrect substitution into the equation given.

- (i) The objective lens magnification of a light microscope can be calculated by:

$$\text{objective lens magnification} = \frac{\text{magnifying power}}{\text{eye piece lens magnification}}$$

Calculate the objective lens magnification of the light microscope with a $\times 10$ eye piece lens magnification and a $\times 40$ magnifying power.

(2)

Show your working.

$$\frac{10}{40} = 0.25$$

$$\text{objective lens magnification} = \times 0.25$$

1(b)(ii) This question was answered well by the majority of learners. Most learners were able to correctly select the maximum resolving power (resolution) of a light microscope.

Question 2

2(a) Some learners were able to correctly select the nucleus as the cell component labelled Y in Figure 3 from the choices given in the multiple-choice item.

2(b) A large variety of responses were submitted for this question, with only a small number being creditworthy. Some learners gave a response suggesting a confusion with nerves.

The most common answers included 'myofilament', 'myosin', 'actin' and 'sarcomere'. Many students provided answers that were too vague, ie, 'muscle fibre', 'tissue', etc.

The following response was awarded one mark. Singular myofibril would be preferred, but learners were not penalised for the plural. It is unclear if the "s" which makes the response plural has been crossed out, but either way this is creditworthy for the one mark.

(b) Figure 4 shows a diagram of a skeletal muscle fibre.

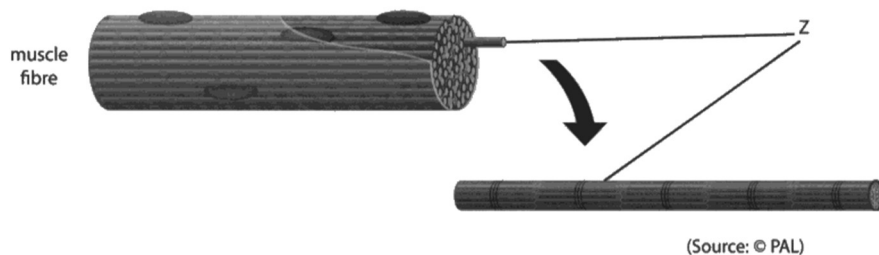


Figure 4

(Source: © PAL)

Identify the structure labelled Z in Figure 4.

(1)

Myofibrils

This response was awarded no marks. It is common incorrect response.

(b) Figure 4 shows a diagram of a skeletal muscle fibre.

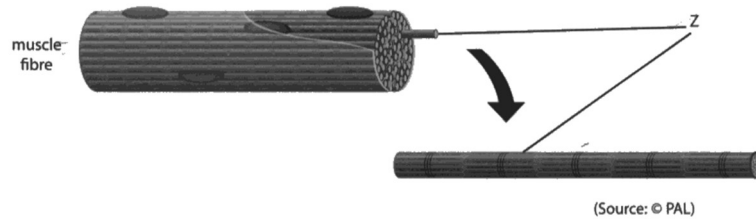


Figure 4

Identify the structure labelled Z in Figure 4.

(1)

Sarcomere

2(c) A significant number of learners did not answer this question. Of those that were able to achieve a mark, many learners scored at least one mark for knowing that the muscle shortened.

Most of the responses were not clear enough to show movement of the Z lines or more overlap of the actin and myosin, with many just drawing a smaller version of the original image. The more able learners either drew the diagram the same size as the original or added arrows to show the movement of the various bits (Z lines moving in) and so gained the marks for demonstrating their knowledge.

The following response was awarded two marks. The Z lines are just closer together and the H-zone is slightly reduced in this example and award mark point 1.

Labels were not required in the learners' image, the components of the diagram are similar to figure 5 could be observed.

There is slightly more overlap of the actin and myosin lines and the overlap seems to increase towards the bottom of the learner's diagram and so therefore mark point 2 would also be awarded.

(2)

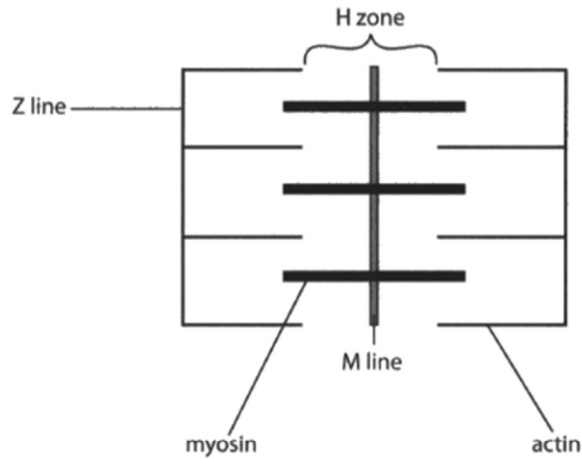
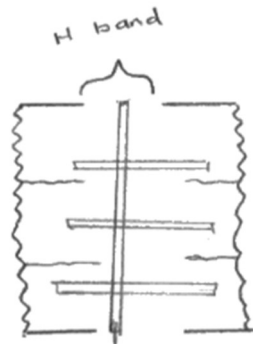


Figure 5



The following response was awarded one mark. The M line has not been drawn in the learner's response, but it was presumed the myosin was represented by the three darker lines (drawn slightly longer but the whole picture is bigger). Therefore, there is more overlap of the actin and myosin- this is mark point 2.

The H zone has been closed up but this has not brought the Z lines closer together. Here the learner almost suggested that the actin stretches, rather than being pulled towards the M line and so mark point 1 cannot be awarded.

One mark was awarded for MP2.

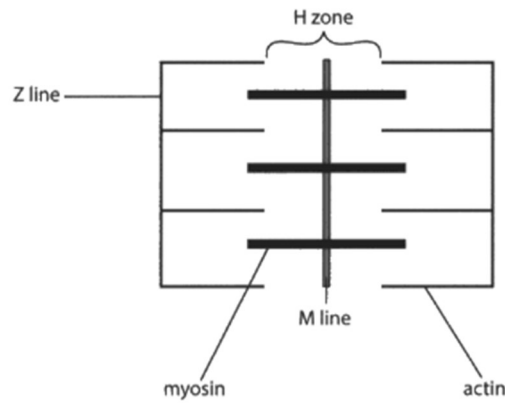
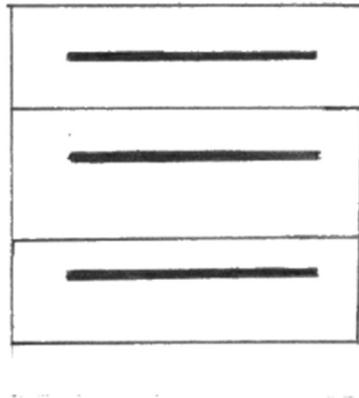


Figure 5



2(d) Learners showed some difficulty in using the correct scientific terms in their answer for this question. A significant number of responses simply repeated the question stem or submitted answers that were too vague to be creditworthy, but most answers scored at least 2 marks.

The most commonly awarded mark was for 'anaerobic respiration', followed by a reference to muscles contracting faster and releasing energy more quickly. It was rare to see a response mentioning ATP in the correct context.

Some learners just compared slow vs fast twitch rather than addressing the question or they described fast twitch muscles. Many learners referred to muscles moving faster/faster movement/movement with more power as opposed to describing the contraction of the muscles. There were lots of references to using up energy quickly, rather than the fast release of energy and very few references to the breaking down / hydrolysis of glucose.

The following response was awarded three marks.

The learner selected MP5 as their identification point (muscles contract faster) any of the points can be the identification point and any can be the expansion points (as long as the response is logical and implies an appropriate order).

The learner has stated that the fast twitch muscle fibres have a higher glycogen content, which is mark point 2. They do not have to say 'stored'. Using glycogen is sufficient for mark point 2.

The learner explains that this gives short burst of ATP, which is acceptable phrasing for mark point 4. The learner has also mentioned fatigue in the right context for the question, which is mark point 7.

For this response, the maximum of three marks was awarded.

A higher percentage ⁽³⁾ fast twitch muscle fibres allow muscles to contract faster than slow twitch muscle fibres due to their higher glycogen content allowing short bursts of ATP to be used, which would give them lots of energy and is why they can run very fast at short distances as they are subject to fatigue if running for a longer period of time.

(Total for Question 2 = 7 marks)

The following response was awarded one mark. The learner has referenced an increased amount of glycogen, which is accepted as mark point 2. The comment about oxygen can be ignored.

They do not get mark point 1, but this does not stop them getting further marks if there are more creditworthy comments after this even though their answer is now in the context of aerobic rather than anaerobic respiration.

The learner explains that the muscle is provided with ATP. "Energy" was allowed in place of ATP, but the learner has not referenced producing the ATP quickly. Lasting a short period of time is not the same as being produced quickly and so wasn't accepted as mark point 4.

(3)

There is an increased amount of glycogen for energy as well as oxygenated blood cells oxygenated that provide the muscle with energy (ATP) that lasts only a short period of time.

(Total for Question 2 = 7 marks)

Question 3

3(a) Some learners were able to correctly select the wall the cell component in Gram-positive bacteria that causes them to appear purple after Gram staining.

3(b)(i) This question was answered well overall, with the majority of learners achieving full marks. The most common mistakes were caused by not understanding the mathematical notation in the given equation and so carrying out the processes in the incorrect order- such as $12 \times (\pi)^2$ or $(12\pi)^2$.

This response was awarded the maximum of two marks.

Learners have been asked to use 3.14 but some have used the value from their calculator for pi. Some learners have rounded their answer and so any answer which rounds to 452, including 452, was accepted for two marks.

If the learner used 22/7 as pi, they will get the answer 452.57 and so we can also award 2 marks for 453 or 452.6.

$$\pi \times 12^2 \quad (2)$$

$$452.16$$

$$\text{area of the zone of inhibition} = 452.16 \text{ mm}^2$$

3(b)(ii) There were very few creditworthy responses for this question. The most common error was stating that Gram positive bacteria were susceptible to penicillin because they lacked the outer structure of the Gram negative bacteria and/or had thin walls that let the penicillin in. Many simply stated that bacteria did not grow where the penicillin is put or that penicillin kills some bacteria. Some learners appeared to confuse the word 'inhibit' with 'inhabit'. Other incorrect responses included reference to bacteria feeding off penicillin, penicillin producing antibodies, penicillin producing Gram negative bacteria to cancel out the Gram positive bacteria, penicillin engulfing

bacteria, penicillin being a healthy bacterium, positive cancelling out negative, penicillin containing white blood cells, and because it's purple.

Responses that were able to achieve a mark did so by referencing the bacteria's thick wall or thick layer of peptidoglycan. Few knew the mechanics of penicillin action inhibiting cell wall synthesis and leading to cell bursting and gave general responses about the antibiotic 'killing bacteria'.

This response was awarded two marks. A great start, but the learner only gives mark point 3 about preventing the cell wall synthesis. The learner has just about enough for mark point 2: the penicillin only acts on dividing/multiplying cells. This is a little more knowledge than "prevents growth" which is just a repeat of the stem.

(iii) Explain why penicillin inhibits the growth of Gram-positive bacteria.

(3)

The gram po penicilan prevents the cell wall synthesis of the gram positive bacteria which makes it sensitive to the penicilan antibiotics. As it is sensitive to the penicilin, it stops them from multiplying.

(Total for Question 3 = 6 marks)

This response was awarded no marks as it was too vague to be creditworthy.

Because penicillin / antibiotics are used to treat Gram-positive bacteria, And therefore creates an environment that the bacteria is unable to survive in.

Question 4

4(a) The most common error submitted for this question was to simply state 'polarisation' or just reading the value from the graph where the label line pointed, rather than identifying the stage that was indicated. Many learners did not appear to know the difference between depolarisation and repolarisation.

This response was awarded one mark. The learner hasn't just mentioned "depolarisation", but has described the action potential as depolarised, which we can accept and award the mark.

(1)
At this stage the action potential is depolarised

This response was awarded no marks. The learner has crossed out the correct answer and given an incorrect answer.

Figure 7

(a) Identify the stage of the action potential that is represented by the ▲ in Figure 7.

(1)
~~Hyperpolarisation~~ Depolarisation Hyperpolarisation

4(b) A large majority of learners were able to give the threshold value shown in Figure 8. A range of -45 to -55 mV was accepted. The most common error was not giving the minus sign.

4(c) A large majority of learners either did not attempt this question or provided a response that was not creditworthy. Some responses displayed a poor understanding of resting potential and wrote about the action potential instead of referring to sodium and calcium channels. Most lost a mark by not saying that -70 mV was inside the axon. Another common error was stating that the membrane is maintained through (voltage gated) ion channels, which rely on diffusion rather than the active transport of ions through pumps.

Others achieved a mark for mention of Na/K pumps, but often got them the wrong way round.

This response was awarded four marks. Marks can be awarded for diagrams. The sketch at the bottom of the response just about shows 3 Na⁺ going out and 2 K⁺ going in through a pump (it is borderline but this is confirmed in the written response) and so mark point 1 and 2 can be awarded. The diagram also shows Cl⁻ inside the membrane, which is mark point 6: this is not seen often but is a marking point. The reference to chlorine in their prose can't be ignored, because they have shown Cl⁻ in their diagram.

The learner has referenced increased permeability to potassium, compared to sodium in their last three words before their diagram.

an axons resting potential -70mv is maintained via the sodium potassium pump. chlorine is already present within the axon membrane. The sodium potassium pump allows 3Na⁺ to diffuse out of the membrane and 2K⁺ to diffuse in. maintaining the resting potential at -70mv the membrane at this point is

The following response was awarded one mark. The details around the Na⁺/K⁺ pump are a little confused. The learner suggests the potassium is pumped in to buffer out the sodium. This is not enough for mark point 1.

The learner has however, stated the inside is negative and this is sufficient for mark point 7.

due to potassium being pumped into the axon the resting membrane is maintained as it buffers out the sodium and keeps the inside negative so that it can be ~~also~~ restarted. It also makes sure that the action potential is ~~at~~ under the threshold. ~~so that~~

(Total for Question 4 = 6 marks)

Question 5

Many of the responses to this question contained answers that were either very generic or did not contain enough information to be creditworthy. Of those responses that were able to score, most achieved three marks or fewer.

Learners were able to discuss the causes of high blood pressure, but many spent a lot of time repeating the stem of the question. The idea that atherosclerosis is simply the deposits from fat in the diet was commonly seen. Mention of LDLs and white blood cells was rare, although some learners did mention cholesterol. There was some confusion about gaseous exchange occurring in arteries. Only a few learners referred to the actual causes of atheroma, eg, inflammation, WBCs, and achieved levels 2 or, even more rare, level 3.

The most common mark points achieved were for reference to the narrowing of lumens and mention of the thickening of the artery wall. Most creditworthy responses referred to atherosclerosis but did not explain the process of how it would have occurred.

The following response was awarded band 3 (six marks). The learner has written a logical, well structure discussion. They have gone a little beyond the scope of the question with the comment about rupture and blood clot, but this is a logical end to their response.

Numbered points or bullet points are completely acceptable they do not have to write in prose. Credit was given to diagrams.

The learner has given a number of points and has formed a logical discussion about how high blood pressure can result in the lining of arteries being damaged. They have brought in ideas from the indicative content, such as foam cells, LDL, endothelial cell, that the deposits are underneath, lumen and then the knock-on effect of increasing the blood pressure further.

- High blood pressure can cause damage to the artery lining because the endothelial cells become damaged due to atherosclerosis.
- Hypertension means that the heart is working harder to pump blood round the body which is why the BP increases.

~~Atherosclerosis~~

Atherosclerosis develops by:

1. The endothelial cells are damaged by ~~to~~ hypertension which then damages the lining of the artery.
2. Foam cells (white blood cells) go towards the damaged artery lining
3. plaque is formed from LDL cholesterol underneath the endothelial cells
4. The lumen diameter narrows → which prevents blood flow
5. blood pressure rises
6. The protective layer of the plaque ruptures
7. A thrombus is formed at the site (blood clot)

- A thrombus would severely damage the artery lining because it prevents a smooth blood flow through the artery.

The following response was awarded band 2 (three marks). Learners have been taught that the artery is narrowed by a plaque and this does support the discussion and so should not be completely ignored.

The learner has given the idea of inflammation and narrow lumen. They have said that smoking can be a cause. There are some confused statements, such

as the red blood cells must work harder, and that the fatty plaque is calcium but there is nothing completely irrelevant.

The learner has repeated some ideas from their first page onto the second page so credit cannot be given for the same idea, just rephrased. They have made an attempt to describe the process of cause and effect.

(6)

High blood pressure can lead
to the build-up of plaque
in the lumen of the arteries
which is typically caused by
inflammation of the blood vessel.
High blood pressure causes an
increase in flow of
blood cells due to
the higher workrate required
in the blood vessels which
can result in inflammation.
People who smoke are

Susceptible to atherosclerosis due to the binding of carbon monoxide to the red blood cells, which ~~may~~ result in a higher blood pressure. As atherosclerosis causes the lumen to narrow, blood cells have to work harder to pass ^{travel} through, which requires a higher blood pressure. In order to occur, the fatty plaque build-up is calcium.

This response was awarded band 1 (one mark). This response contains an isolated piece of knowledge (inflamed) and therefore is awarded one mark as it does not fulfil the "no awardable content" descriptor for no marks.

The start of the response is a little vague and seems to be about the heart rather than the arteries. Had they given a little more detail about the strain on the heart, this answer could have almost reached the top of band 1 as ideas about the causes of the damage, including lifestyle factors, could be credited. However, the learner is not entirely clear here.

(0)

Because of high blood pressure the heart is constantly pumping ^{meaning} ~~until~~ that the walls of tissue are being overworked, causing them to become inflamed.

Chemistry

Question 1

This question of the paper focused on the element of period 3.

Q1(a)(ii) asked learners to give the symbol of one of the elements in period 3 that can form covalent bonds with oxygen.

Some learners interacted well with the question and stated that Si, P, S or Cl, as in this example that scored one mark.

(b) Give the symbol of **one** of the elements in period 3 that can form covalent bonds with oxygen.

(1)

Cl

Unfortunately, a large proportion of learners did not read the question carefully and gave the name of any element that would form a bond with oxygen, such as magnesium, rather than giving an element that would form a covalent bond.

(b) Give the symbol of **one** of the elements in period 3 that can form covalent bonds with oxygen.

(1)

Mg

Q1(c) asked learners to give the symbols of the two elements in the s-block of the periodic table. A good proportion of learners were able to give the symbols of the elements and gained both marks. This example scored two marks.

(c) Give the symbols of **two** elements in period 3 that are in the s-block of the periodic table.

(2)

Na and Mg

Some learners used the s-block and gave elements in period 2 rather than period 3.

(c) Give the symbols of **two** elements in period 3 that are in the s-block of the periodic table.

(2)

Li and Be

Q1(d) Learners found this question difficult with the majority giving the symbol for argon rather than chlorine. This example scored no marks.

(d) Give the symbol of the most electronegative element in period 3.

(1)

Ar

Q1(e) was the last on the elements on period 3 and asked learners to explain the trend in melting point from sodium to aluminium. Many learners were able to score at least one mark for identifying the trend of increasing melting point.

It was clear that few learners knew how to explain why the melting point increases with few linking their answer to the structure of metals, positive ions, delocalised electrons or metallic bonding. Marks were often lost as learners described intermolecular forces rather than electrostatic forces between the (positive) ions or nuclei and delocalised electrons or the metallic bonding.

In this example, one mark was gained for giving the trend in melting point. Unfortunately, the learner then refers to intermolecular forces rather than electrostatic attractions.

(e) Explain the trend in melting point from sodium to aluminium.

(4)

The Melting Point for Sodium ~~increases~~ ^{is} lower than of Aluminium. Between Sodium and aluminium the melting point continues to increase as the intermolecular bonds at the elements become stronger therefore more energy is required to overcome them.

This example shows a good response that related this structure of the metals to the stronger electrostatic attraction and scored all four marks available.

(e) Explain the trend in melting point from sodium to aluminium.

(4)

Sodium, magnesium and aluminium all form metallic structures. The atoms lose their outer electrons to produce positive metal ions with a sea of delocalised electrons. The more electrons are donated to the sea, the more electrostatic attraction between the ions and electrons and the more energy required to break these bonds. Sodium loses one electron to form an ion, magnesium two and aluminium three. Therefore the melting point increases from sodium to aluminium.

Question 2

This question focused on ammonia.

Learners found it difficult to draw the hydrogen bond between the two molecules of ammonia. A common error was to draw an extra hydrogen bonded in-between the N and H as in this example which scored no marks.

2 Ammonia, NH_3 , is a simple molecular compound.

(a) Hydrogen bonding occurs between molecules of ammonia.

Figure 2 shows two molecules of ammonia.

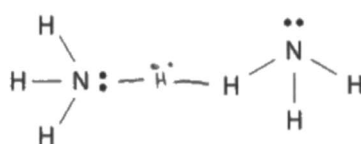


Figure 2

Draw **one** hydrogen bond between the two molecules shown in Figure 2.

You should include the partial charges on the molecules.

(2)

In some cases, learners gave charges on parts of the molecules, but these were often in the wrong places or were not partial charges but just a + or – which were not accepted.

2 Ammonia, NH_3 , is a simple molecular compound.

(a) Hydrogen bonding occurs between molecules of ammonia.

Figure 2 shows two molecules of ammonia.

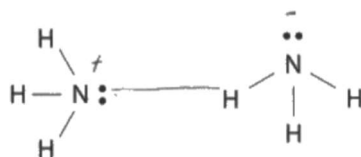


Figure 2

Draw **one** hydrogen bond between the two molecules shown in Figure 2.

You should include the partial charges on the molecules.

(2)

This last example scored full marks for the partial charges and the representation of the hydrogen bond.

2 Ammonia, NH_3 , is a simple molecular compound.

(a) Hydrogen bonding occurs between molecules of ammonia.

Figure 2 shows two molecules of ammonia.

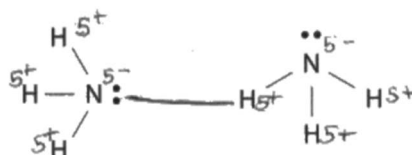


Figure 2

Draw **one** hydrogen bond between the two molecules shown in Figure 2.

You should include the partial charges on the molecules.

(2)

2(b)(ii) asked learners to give the definition of a dative covalent bond. Few learners were able to the correct definition with a common misconception being that it was a like a covalent bond but with two or more bonds.

Where marks were awarded, it was often for stating that there was a shared pair of electrons, with few of those stating that both of the electrons come from the same atom. In some cases learners were close and stated that electrons were shared but did not say how many.

This example scored one mark for stating that is shared pair of electrons, there was no reference to both of the electrons coming from the same atom for the second marking point.

(ii) The ammonium ion contains a dative covalent bond.

Describe what is meant by a **dative covalent bond**.

(2)

A shared pair of electrons

A good example that scored both marks.

(ii) The ammonium ion contains a dative covalent bond.

Describe what is meant by a **dative covalent bond**.

(2)

A dative covalent bond is a bond where the shared pair of electrons comes from the same atom.

Q2(c)(i) asked learners to consider the reaction of nitric acid with ammonia to form ammonium nitrate and to find the limiting reagent in the reaction. A good proportion of learners were able to gain the first mark for calculating the number of moles that were present of each reagent. Some of these then took this forward to put nitric acid as the limiting reagent on the answer line. Some learners put the formula for nitric acid on the answer line, this was also accepted for the second marking point and this example scored both marks.

(c) (i) Ammonium nitrate is another salt of ammonia that can be used as a fertiliser.

It can be made from the reaction of ammonia with nitric acid.

The balanced equation for the reaction is

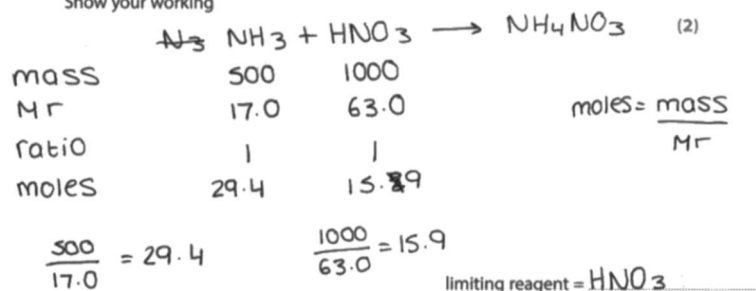


A manufacturer reacts 500 g of ammonia with 1000 g of nitric acid.

Determine the limiting reagent in the reaction.

(relative formula masses $\text{NH}_3 = 17.0$, $\text{HNO}_3 = 63.0$)

Show your working

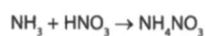


However, a noticeable proportion then carried out a further, incorrect calculation with the two numbers including, dividing, multiplying or subtracting the numbers. Showing a lack of understanding of the term limiting.

(c) (i) Ammonium nitrate is another salt of ammonia that can be used as a fertiliser.

It can be made from the reaction of ammonia with nitric acid.

The balanced equation for the reaction is



A manufacturer reacts 500 g of ammonia with 1000 g of nitric acid.

Determine the limiting reagent in the reaction.

(relative formula masses $\text{NH}_3 = 17.0$, $\text{HNO}_3 = 63.0$)

Show your working ammonia nitric acid

(2)

$$\begin{aligned} \frac{500}{17.0} &= 29.41176471 \\ &= 13.53 \\ \frac{1000}{63.0} &= 15.87301587 \\ &\div = 1.852 \end{aligned}$$

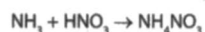
limiting reagent = 13.53

In some cases, the learners inverted their calculation for the number of moles and so did not score marking point 1. However, error carried forward was allowed and so in this case, credit was awarded for ammonia for marking point 2 and so one mark was awarded.

(c) (i) Ammonium nitrate is another salt of ammonia that can be used as a fertiliser.

It can be made from the reaction of ammonia with nitric acid.

The balanced equation for the reaction is



A manufacturer reacts 500 g of ammonia with 1000 g of nitric acid.

Determine the limiting reagent in the reaction.

(relative formula masses $\text{NH}_3 = 17.0$, $\text{HNO}_3 = 63.0$)

Show your working

(2)

$$\begin{aligned} \frac{17}{500} &= 0.034 \\ \frac{63}{1000} &= 0.063 \end{aligned}$$

limiting reagent = NH_3

Q2(c)(ii) Learners were asked to calculate the mass of ammonia in a solution in grams.

Learners found it difficult to score all three marks on this question, however a good proportion picked up at least one mark, this was either scored for the conversion of the volume to dm^3 or for multiply the concentration by the relative formula mass.

A good, well set out answer that scored all three marks.

- (ii) A student wants to carry out the same reaction in the laboratory on a smaller scale.

The student uses 50 cm^3 of a 0.3 mol dm^{-3} solution of ammonia.

Calculate the mass of ammonia in the solution in grams.

(relative formula mass $\text{NH}_3 = 17.0$)

Show your working

$$\text{moles of NH}_3 = \frac{\text{mass}}{\text{Mr}}$$

(3)

$$1\text{ dm}^3 = 1000\text{ cm}^3$$

$$50\text{ cm}^3 = 0.05\text{ dm}^3$$

$$\text{moles} = 0.3 \times 0.05 = 0.015\text{ moles}$$

$$\text{mass} = 0.015 \times 17 = 0.255\text{ g}$$

$$\text{mass of ammonia} = 0.255\text{ g}$$

In this example, the learner has multiplied the relative concentration by the volume to obtain the number of moles and multiplied by the relative formula to get the mass. Unfortunately, they have not converted the volume to dm^3 and so missed out on the first marking point. The answer scored two marks.

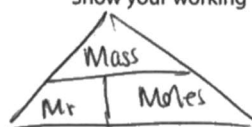
- (ii) A student wants to carry out the same reaction in the laboratory on a smaller scale.

The student uses 50 cm^3 of a 0.3 mol dm^{-3} solution of ammonia.

Calculate the mass of ammonia in the solution in grams.

(relative formula mass $\text{NH}_3 = 17.0$)

Show your working



$$\text{Mr} = 17.0$$

$$\text{Moles} = 0.3$$

$$17.0 \times 0.3 \times 50\text{ cm}^3 = 255\text{ grams}^{(3)}$$

$$\text{mass of ammonia} = 255\text{ g}$$

In this next example, the learner has converted the volume into dm³ and then multiplied by the concentration to get the number of moles, unfortunately they did not multiply by the relative formula mass to obtain the mass in grams as required by the question and so only scored two marks.

- (ii) A student wants to carry out the same reaction in the laboratory on a smaller scale.

The student uses 50 cm³ of a 0.3 mol dm⁻³ solution of ammonia.

Calculate the mass of ammonia in the solution in grams.

(relative formula mass NH₃ = 17.0)

Show your working

$$\text{moles} = \text{mass} \div \text{volume} \quad \text{or} \quad \text{moles} = \frac{\text{mass}}{\text{Mr}}$$

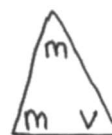
$$\text{moles} = 0.3$$

$$\text{volume} = 0.05$$

$$\text{mass} = ?$$

$$\frac{50}{1000} = 0.05$$

$$0.3 \times 0.05 = 0.015 \text{ g}$$



(3)

$$\text{mass of ammonia} = 0.015 \text{ g}$$

Question 3

This question focused on the relative size of atoms and ions across period 2.

The first question was a multiple choice item asking learners to describe how the atomic radius is expected to change across the period. Many thought that the atomic radius would decrease and then increase across the period and put answer C rather than the correct answer, A, that the atomic radius would decrease across the period.

Q3(b) learners were given the electronic configuration of an oxygen atom and then asked to complete the electronic configuration of an oxide ion, O^{2-} . A common misconception was that the atom had lost two electrons and therefore a common answer seen was $1s^2 2s^2 2p^2$, as in this example that did not score.

(b) The electronic configuration of an oxygen atom is $1s^2 2s^2 2p^4$.

Complete the electronic configuration of the oxide ion, O^{2-} .

(1)
 $1s^2 2s^2 2p^2$

The following example scored the mark.

(b) The electronic configuration of an oxygen atom is $1s^2 2s^2 2p^4$.

Complete the electronic configuration of the oxide ion, O^{2-} .

(1)
 $1s^2 2s^2 2p^6$

Q3(c) Learners were asked to explain the trend in the size of the ionic radius from lithium to Boron. Whilst a good proportion scored for identifying the trend, few were able to take this further to explain that whilst each positive ion had the same number of electrons because the number of protons was increasing there is a greater electrostatic attraction between the electrons and protons which is why the ionic radius decreases.

This next example scored just one mark for identifying the trend. Common misconceptions were to again focus on intermolecular forces or to discuss ionisation energy or electronegativity.

(c) Explain the trend in the size of the ionic radius from Li^+ to B^{3+} .

(3)

the ionic radius shrinks as there starts to be less ions in them.
the ionic radius shrinks as there starts to be less electrons in the outer shells
of these ionic compounds

This example scored the full three marks available.

(c) Explain the trend in the size of the ionic radius from Li^+ to B^{3+} .

(3)

The overall trend is that the ionic radius from Li^+ to B^{3+} decreases as ~~there are~~ there are more protons in the positive nuclei. This increasing number of protons means there is a stronger force of attraction so the electrons (negative) are pulled towards the nuclei (positive) causing a reduction in ionic radius.

(Total for Question 3 = 5 marks)

Question 4

This question gave learners a graph of the ionisation energy for each successive electron removed from an atom of silicon. They were then asked to explain how the pattern shown in the graph could be used to determine the group and period of the element.

The most common error was misreading the graph. Many students discussed the change in ionization energy across a period or down a group rather than relating the graph to silicon. Many students also started by stating what ionization energy is which did not answer the question.

In the next example, the learner has described how to work out the group number and given group 2 as an example, but they have not actually linked this to silicon.

The learner has also described how to work out the change in period number and shown this on the graph. However, these ideas have not been linked to silicon and there is no real explanation of why this happens, this therefore gained a mark of two in level 1.

from an atom of silicon.

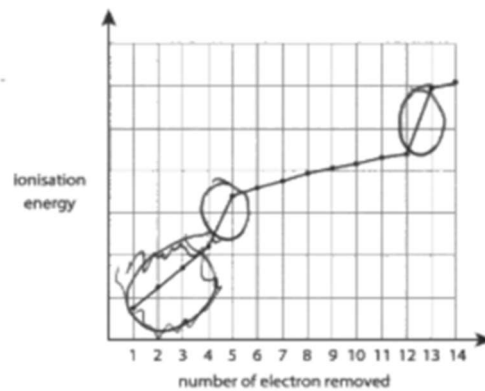


Figure 4

Explain how the pattern of ionisation energies shown in Figure 4 can be used to determine the group and period of the element.

(6)

As the number of electrons removed increases, ~~so do~~ the ionisation energy increases. You can determine the group of the element by observing the number of electrons removed e.g. 2 electrons removed = group 2. You know this because the number of electrons removed pertains to the number of electrons in the outer shell.

To work out period you look for the large change in ionisation energy, e.g. circled above. This indicates that a new period is being recorded.

In this example, the learner has given a good explanation as to what group silicon can be found in, including reference to the amount of energy required to remove the electrons as they were in the outer shell and further from the nucleus, the fact that the jump then means that you are in another shell so closer to the nucleus and less shielding. There is no explanation of the period number and so a mark at the top of level 2 of four was awarded.

4 Figure 4 shows a graph of ionisation energy for each successive electron removed from an atom of silicon.

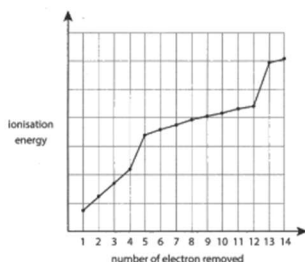


Figure 4

Explain how the pattern of ionisation energies shown in Figure 4 can be used to determine the group and period of the element.

(6)

The pattern reveals that the element is in group four because there is a steady increase for the first four electrons, and then there is a large increase ~~to~~ between the fourth and the fifth. This means that there were four electrons in the outer shell of this element because they ~~is~~ required less energy to remove them as they are further from the nucleus. The large increase shows a change in shells because the electrons are closer to the nucleus and experience less shielding so it requires more energy to remove one. There being four electrons in the outer shell means that the element is group four because group four

elements have four electrons in their outermost shell. The element must be in period 3 because 14 electrons were removed which is the number of ~~pro~~ electrons in silicon which is in group 3

(Total for Question 4 = 6 marks)

TOTAL FOR SECTION B = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS

In this final example, the learner starts with some good annotations on the graph which shows a good understanding and adds to the answer given.

They have given a good discussion of why silicon is in group 4 in the 1st paragraph and go on to explain in the 2nd paragraph why it is in period 3. Within the discussions, they have added in deeper discussion in terms of the 1st shell requiring the most energy to remove because it is closest to the nucleus and the last shell requiring less energy because it is further away from the positive nucleus. Although there is no discussion of shielding or electrostatic attraction, this was considered sufficient for full credit at level 3 and six marks awarded.

4 Figure 4 shows a graph of ionisation energy for each successive electron removed from an atom of silicon.

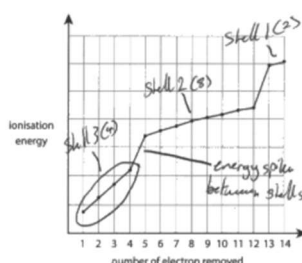


Figure 4

Explain how the pattern of ionisation energies shown in Figure 4 can be used to determine the group and period of the element.

(6)

The graph shows to us how many electrons the atom has to lose, which would show us that this is an atom of Silicon as it only has 14 electrons. This graph can show what group the atom is in by showing the first 4 electrons removed take less energy than the other electrons therefore showing how this is the outer shell and then 4 electrons are the valent electrons therefore this atom is in group 4.

This graph shows how the atom is in period 3, this is due to the atom having 2 clear spikes in energy required to remove the next shell of electrons as has been labelled on Figure 4. What this tells us is that there are three shells, the first shell of the atom requires the most energy to remove an electron as it's closest to

the nucleus and has 2 electrons, the second shell has 8 electrons and requires less energy than shell 1 due to it being further away from the positive nucleus. And shell 3 has 4 electrons so it is incomplete and requires even less energy than the first two shells as it is further away from the nucleus. We also get the electron configuration due to identifying the separate shells and number of electrons in them, the electron configuration is $2, 8, 4$ which you can use to find silicon on the periodic table.

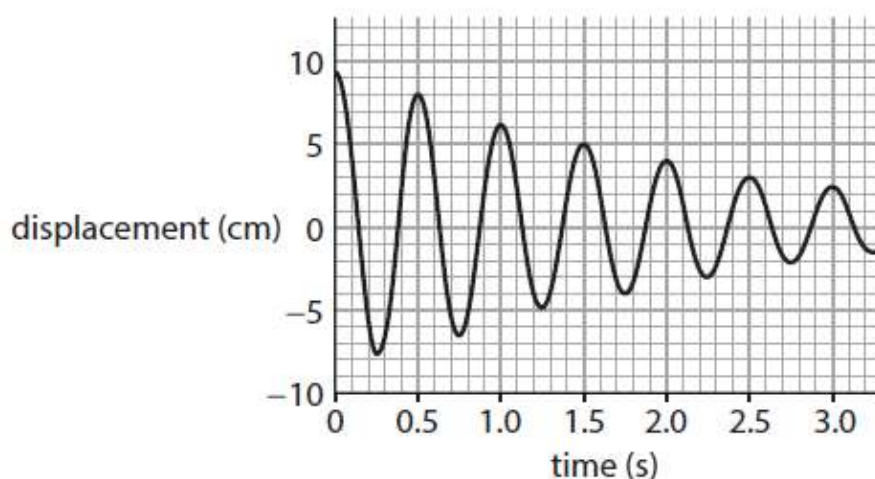
(Total for Question 4 = 6 marks)

TOTAL FOR SECTION 8 = 30 MARKS
TOTAL FOR EXAMINATION = 90 MARKS

Physics

Question 1

Question 1(a)(i) and 1(a)(ii) refer to the same diagram.



1(a)(i) The majority of learners recalled that the amplitude of a wave is the displacement taken from the rest position and correctly gave the amplitude at the time of two seconds to be exactly 4cm.

1(a)(ii) About 60% of learners were able to determine the periodic time (T) of the oscillation, this is 0.5 seconds being the time between two successive peaks on the graph.

1a(iii) The majority of learners were able to complete the calculation using the equation $f = 1/T$. The value of the period T had to be taken from part 1(a)(ii) of the question.

(ii) Give the periodic time (T) for one complete oscillation. (1)

time for one oscillation (T) = 0.5 s

(iii) Calculate the frequency (f) of the oscillations.

Use the equation: $f = \frac{1}{T}$

(2)

Show your working.

$$1/0.5 =$$

frequency of oscillations (f) = 2 Hz

1(b) This question was quite difficult for all learners because the information on superposition of waves was given in two ways. Most learners would know the superposition was destructive but relating this to a phase difference of 180° was more difficult.

(b) Figure 3 shows two waves which are to be superposed.

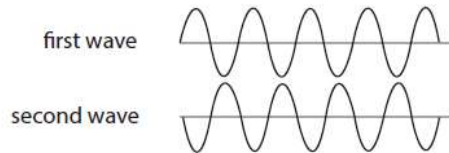


Figure 3

Which row A, B, C or D of the table correctly identifies the type of superposition and the phase difference between the waves.

(1)

		type of superposition	phase difference between waves
☐	A	constructive	180°
☐	B	constructive	360°
☐	C	destructive	180°
☐	D	destructive	360°

Question 2

The question included both longitudinal and transverse waves.

2(a) This required learners to recall the names used to describe the parts of a longitudinal wave passing along a spring. Only about half of learners were able to recall these correctly.

2 Longitudinal and transverse are two types of wave.

(a) Figure 4 shows a longitudinal wave produced on a long spring.

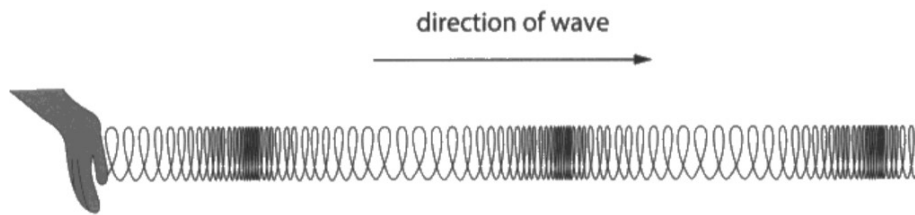


Figure 4

Complete Sentence 1 in the box to describe the longitudinal wave in Figure 4.

The longitudinal wave travels along the spring as a series

of compressions and rarefactions

2(b) About half of learners were able to give a correct answer when asked for a use of infrared waves. It was expected that learners knew the uses of the waves that form part of the electromagnetic spectrum. The most common correct answer was TV remote control although the use in fibre optic communication, night vision cameras was sometimes given.

(b) Infrared waves are an example of transverse waves.

Give **one** use of infrared waves in communication.

(1)

TV remote

(b) Infrared waves are an example of transverse waves.

Give **one** use of infrared waves in communication.

(1)

Heat sensor cameras.

2(c) The diagram shows a transverse wave and about half of learners were able to gain one mark for describing the movement at P but relating this motion up and down to the direction of travel of the wave proved to be difficult for many learners. The following responses gained both marks:

(c) Figure 5 shows a transverse wave being produced using a long spring.

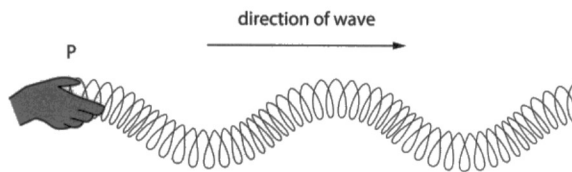


Figure 5

Describe how the movement of the spring at P produces a transverse wave.

(2)

The wave is curved and the oscillations are perpendicular to the energy transfer.

Describe how the movement of the spring at P produces a transverse wave.

(2)

The Spring is being moved up and down which creates waves which are ~~parallel~~ perpendicular to the direction of travel.

Question 3

3(a) Learners found this question very difficult. They were able to recognise and give features of a stationary wave but were not able to describe, with the apparatus given, how the stationary wave was produced. The wave is produced by the oscillator, travels down the string and is reflected so the waves superimpose to give nodes and antinodes.

The following answer covers all four marking points in the mark scheme:

- 3 Figure 6 shows a string under tension. An oscillator is vibrating the string to produce a stationary wave.

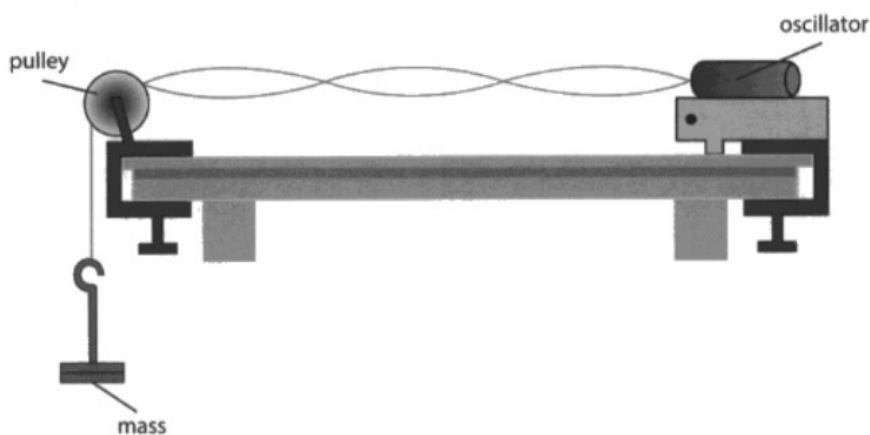


Figure 6

- (a) Describe how a stationary wave is produced in figure 6 using the equipment shown.

(3)

When an incident wave travel towards ~~to~~ a boundary, it reflects and a reflected wave is ~~is~~ formed. This means that there are ~~2~~ two waves at the same place and time travelling in ~~the~~ opposite direction. This means that the waves are ~~a~~ ^{at} superposition and they constructively ^(in phase) and destructively ^(out of phase) interfere at antinodes and nodes.

This answer gives marking points 1, 2 and 4:

(a) Describe how a stationary wave is produced in figure 6 using the equipment shown.

(3)

- String is fixed at 2 points.
- Incident wave is sent down the string, hits the fixed end so a reflected wave is sent back.
- Both waves have the same frequency and wavelength.
- Nodes and anti-nodes are formed.
- Stationary waves are produced by resonance.

3(b) This was a multiple choice question using Figure 6. Learners were expected to identify that the diagram showed the third harmonic on the string. About half of learners were able to identify the third harmonic.

3(c) A third of learners were able to receive full marks and about 80% of learners were able to obtain at least one mark. This calculation uses a familiar equation but the equation is best squared and rearranged to calculate the value of the tension in the string.

The answer below shows this method and that standard form is used with a calculator to give the final answer:

(c) The string has a mass per unit length (μ) of $6.6 \times 10^{-4} \text{ kg m}^{-1}$.

The velocity (v) of the wave on the string is 32.1 m s^{-1} .

Calculate the tension (T) in the string.

(4)

Use the equation: $v = \sqrt{\frac{T}{\mu}}$

Show your working.



$$32.1 = \sqrt{\frac{T}{6.6 \times 10^{-4}}}$$

$$32.1^2 = \frac{T}{6.6 \times 10^{-4}}$$

$$32.1^2 \times 6.6 \times 10^{-4} = 0.6800706$$

tension (T) in the spring = 0.68 N

Not squaring both sides of the equation to remove the square root gave an answer of 0.021N, which would have gained three marks as only one of the marking points had been missed.

It should be remembered that one mark was given for correct substitution into the given equation and that this should have been shown before there was any attempt to rearrange the equation, so that the substitution can be identified.

Calculate the tension (T) in the string.

(4)

Use the equation: $v = \sqrt{\frac{T}{\mu}}$

Show your working.

$$32.1 \text{ m s}^{-1} = \sqrt{\frac{T?}{6.6 \times 10^{-4} \text{ kg m}^{-1}}}$$

tension (T) in the spring = N

Question 4

4(a) This was a multiple choice question and required recall of parts of the electromagnetic spectrum in order of increasing frequency. The order of waves in the electromagnetic spectrum needs to be learned.

Answer: radiowaves, microwaves, infrared.

4(b) Many learners found this question difficult their knowledge of transmission of mobile phone signals was limited. The answer below shows an excellent knowledge of the system and gives a succinct description of the process.

(b) Mobile phones use electromagnetic waves for communication.

Describe how a signal is transmitted from one mobile phone to another mobile phone.

(3)

The signal by a mobile phone is sent to the nearest cell tower through microwaves. The cell tower then sends the signal to the phone centre where the signal is then sent to the cell tower that is closest to the receiving phone allowing it to reach the phone.

The answer above applies to a terrestrial link and the answer below to a satellite phone link both are equally valid but cannot be mixed.

(b) Mobile phones use electromagnetic waves for communication.

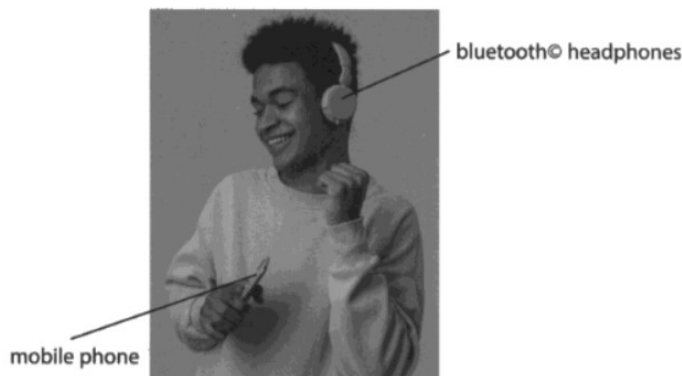
Describe how a signal is transmitted from one mobile phone to another mobile phone.

(3)

mobile phones use microwaves for communication. when a ~~signal~~ a signal is being transmitted from the ground it goes up into space. This is called the uplink frequency. Then it goes to the transponder which is in space that changes the frequency and filters out any interference. Then it goes down to earth to the mobile phone that is being sent to. This is called downlink frequency.

4(c) Learners also found this question difficult with just about half of learners gaining both marks. The difficulty was that learners did not recognise that Bluetooth® is used in low power devices and therefore has a limited range. Two marks were also awarded for relating the strength of the signal to the distance.

(c) A student uses Bluetooth® headphones to listen to music.



(Source ©PAL)

Figure 7

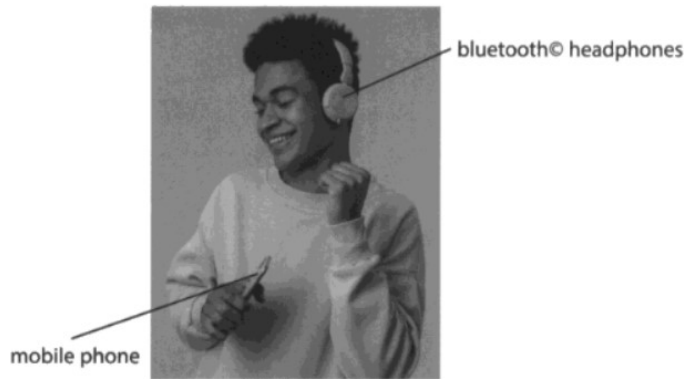
Explain why the Bluetooth® headphones must be near to the mobile phone so that the student can listen to the music.

(2)

Bluetooth is an incredibly weak radio wave being unable to travel long distances and barely make it through walls. This is why when using bluetooth both devices must be close to each other.

(Total for Question 4 = 6 marks)

(c) A student uses Bluetooth® headphones to listen to music.



(Source ©PAL)

Figure 7

Explain why the Bluetooth® headphones must be near to the mobile phone so that the student can listen to the music.

(2)

Bluetooth signals are weak and cannot travel far because the signal will weaken over the distance. So by being close to the phone, the distance is shortened and still able to carry the sound.

Question 5

The majority of learners were only able to achieve level 1 by considering reflection being responsible for the passage of light through the optical fibres or some idea of the structure of the endoscope. This demonstrated some knowledge of scientific facts. An explanation of when total internal reflection occurs or that light has to go up and down the fibres and/or some idea of structure would achieve level 2 as this demonstrated good knowledge and gave a mostly clear explanation. However, to achieve level 3, a detailed explanation of total internal reflection and the idea of incoherent and coherent bundles of optical fibres being used, and the that each fibre of the coherent bundle produced a pixel to make up the image on the screen was needed. This demonstrated comprehensive knowledge and understanding and an explanation which was logical and had a well developed structure.

Common misconceptions were that light was reflected inside the optical fibres by mirrors and that at the end of the endoscope in the stomach there was a camera to take picture.

Level 1 (two marks): gives an idea of the structure of the endoscope without any detail.

- 5 Figure 8 shows an endoscope being used to observe the inside of the stomach of a patient.

Endoscopes use optical fibres.

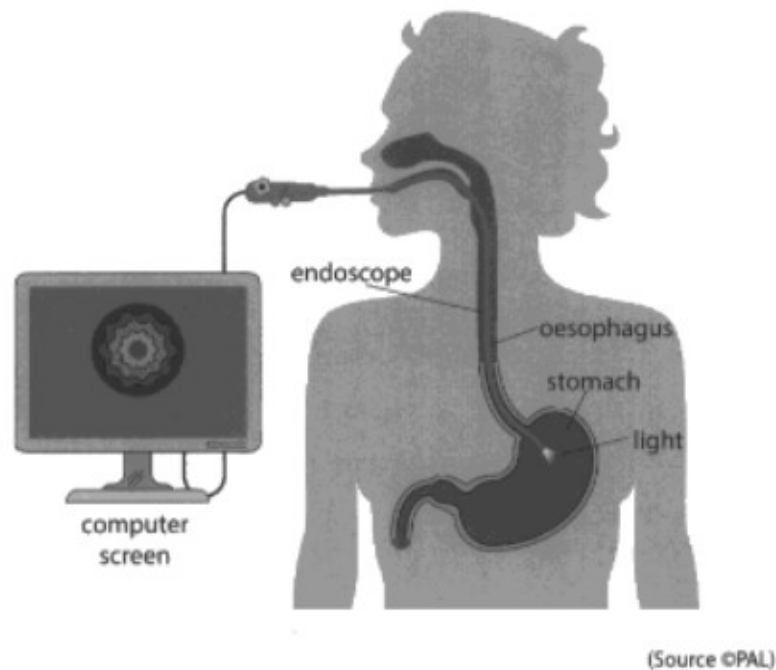


Figure 8

Explain how the optical fibres in an endoscope are used to produce an image on the computer screen.

(6)

You may add a diagram to support your answer.

First a signal is sent through the endoscope into the stomach which bounces back into the endoscope to the computer making a image. The ray signal carries that information through.

Level 2 (four marks): this response has the idea of reflection within the fibre and an idea of the structure in that the light goes down to the stomach and then goes back up to the computer screen. There is also a diagram showing total internal reflection.

- 5 Figure 8 shows an endoscope being used to observe the inside of the stomach of a patient.

Endoscopes use optical fibres.

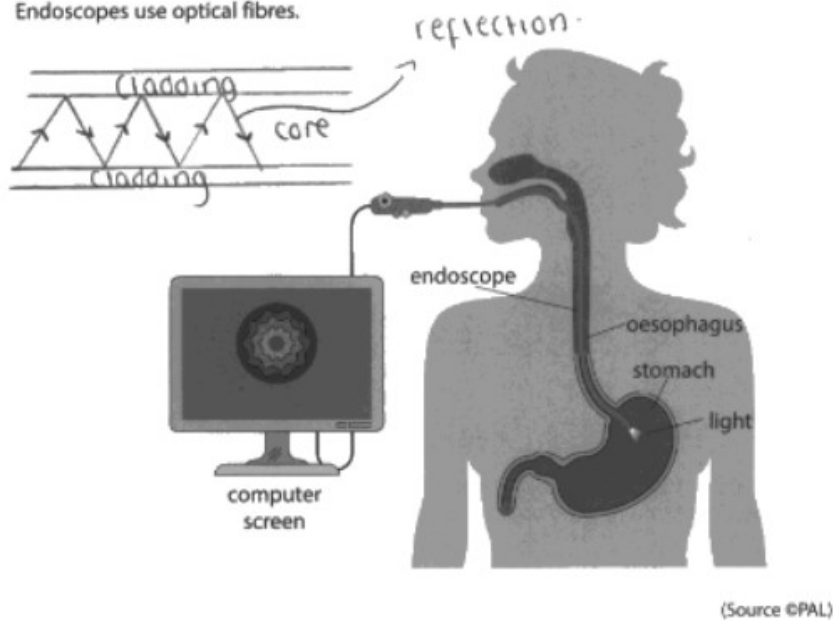


Figure 8

Explain how the optical fibres in an endoscope are used to produce an image on the computer screen.

(6)

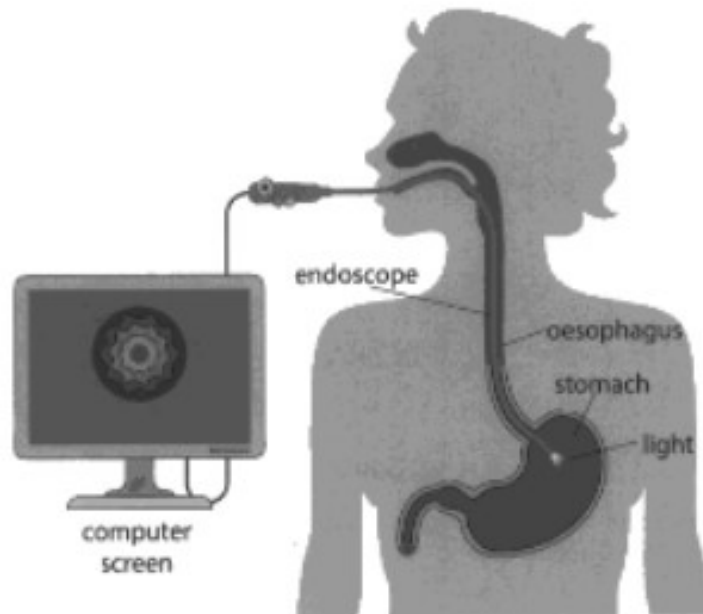
You may add a diagram to support your answer.

- The light is shone through the optical fibres.
- This then reflects off the cladding staying within the optical fibre.
- As it gets towards the end of the optical fibre the light then exits ^{*} ~~for~~ causing the reflection to discontinue ^{*} ~~refracts~~ (refracts).
- This produces an image as it sends the signal back up to the computer screen through the optical fibres by reflecting off the cladding back to the source.

Level 3 (six marks): gives detail of passage of light through the optical fibres and explains how the endoscope works to produce an image on the screen.

- 5 Figure 8 shows an endoscope being used to observe the inside of the stomach of a patient.

Endoscopes use optical fibres.



(Source: OPAL)

Figure 8

Explain how the optical fibres in an endoscope are used to produce an image on the computer screen.

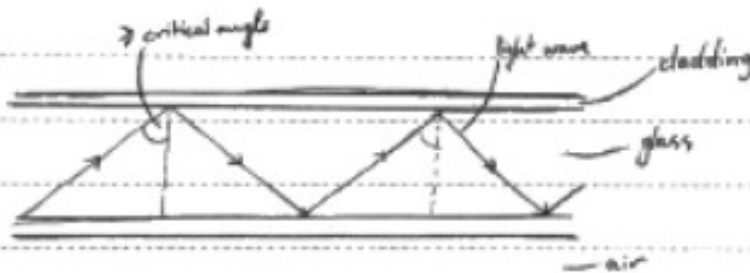
(6)

You may add a diagram to support your answer.

The endoscope sends the light waves through cladded optical fibres at an angle equal to or more than the critical angle for light in that material (glass). This causes total internal reflection, so no light can escape the fibre until it reaches the end. Cladding ensures that no light escapes the fibre, as it has a critical angle smaller than that of the optical fibre due to higher optical density. When the light waves reach the end of the endoscope, they are projected onto the inner surface of the stomach. They reflect off these surfaces to different angles, and the reflected waves travel

back up a separate bundle of optical fibres. These fibres also have the same properties for total internal reflection to occur. Each fibre in this bundle, when the waves have returned back to the top of the endoscope, will be used to piece together an image of the inside of the stomach. The. Each optical fibre ~~is~~ carries information with around 2 pixels on the computer screen.

Optical Fibre



Summary

Biology

Centres need to fully prepare learners for the exam by practicing exam techniques, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question, to re-read their response in order to ensure that the question set has been addressed, and that they have used appropriate scientific knowledge and vocabulary. Learners should have a clear understanding of the difference between “function” and “structure”.

Based on their performance on this paper, learners should:

- Understand the demand of the command verbs, especially the difference between describe and explain.
- Be familiar with and recognise the ultrastructure and function of organelles in the following cells: prokaryote cells (bacterial cells), eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific).
- Revise key definitions with Level 3 detail.
- Show their working out for all calculations to ensure marks can be given for the stages within the calculation.
- Use the additional guidance to understand the depth and breadth of the specification.

Chemistry

Based on their performance on this paper, learners should:

- Ensure they read the questions carefully, consider highlighting key terms and/or command words to enable them to focus on the question in hand.
- Re-read their answers when they have finished to check that the answer they have given answers the question set rather than a question that they think might be there.
- Use past papers to practice exam technique.

- Practice laying out calculations and showing their working in a clear and logical way.
- Avoid putting everything they know about a subject in an answer, try to be concise and only give detail that is relevant to the question posed.

Physics

Based on their performance on this paper, learners should:

- Learn the order in either increasing frequency or decreasing wavelength of waves in the electromagnetic spectrum.
- Learn the uses and applications of waves in the electromagnetic spectrum.
- Be able to describe longitudinal and transverse waves.
- Always show substitution into the equation given at the start of a calculation.
- Be able to use standard form on a calculator.
- Be able to identify phase difference of waves in terms of angle.
- Learn the difference between refraction and reflection.



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